

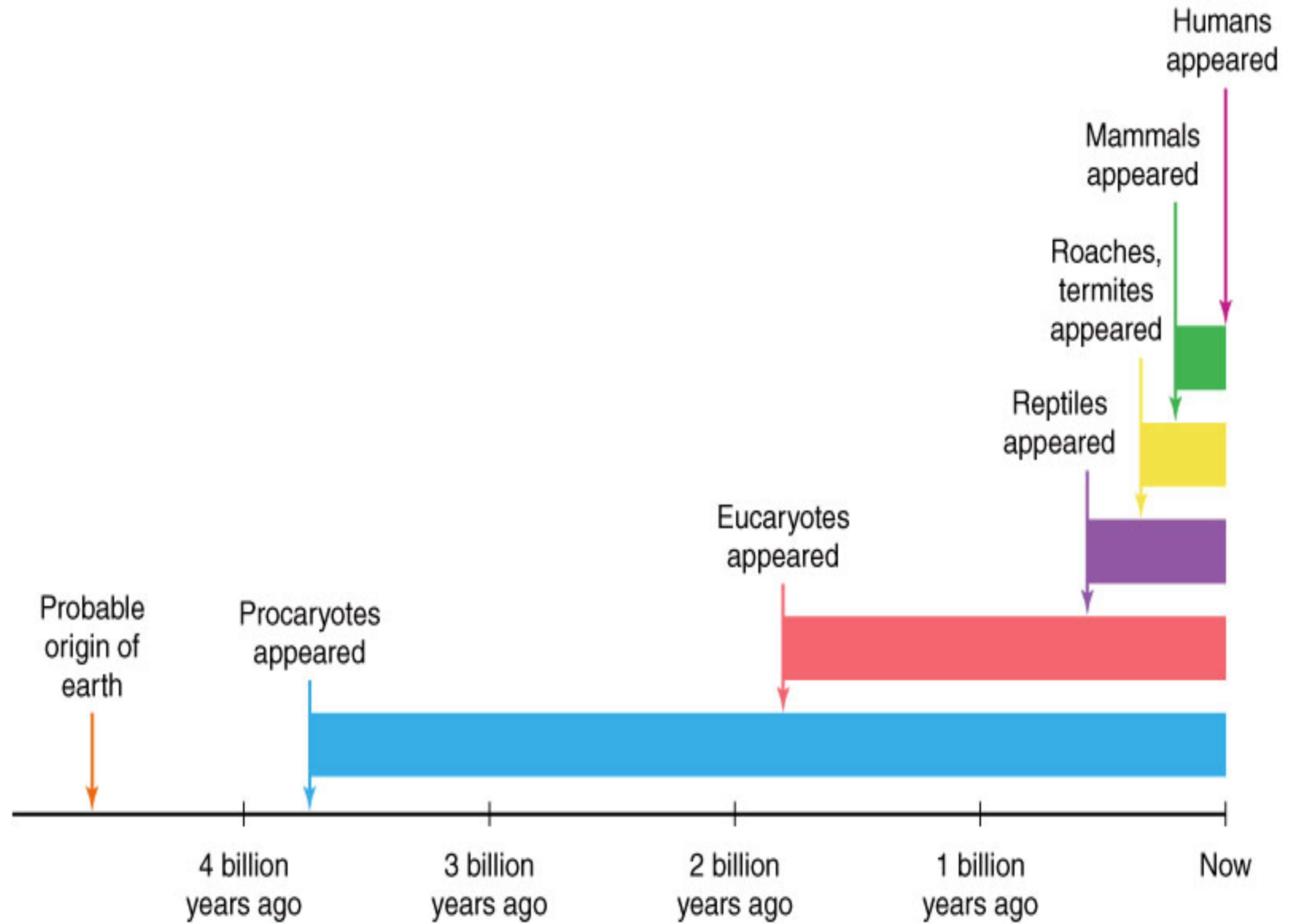
Lecture 6

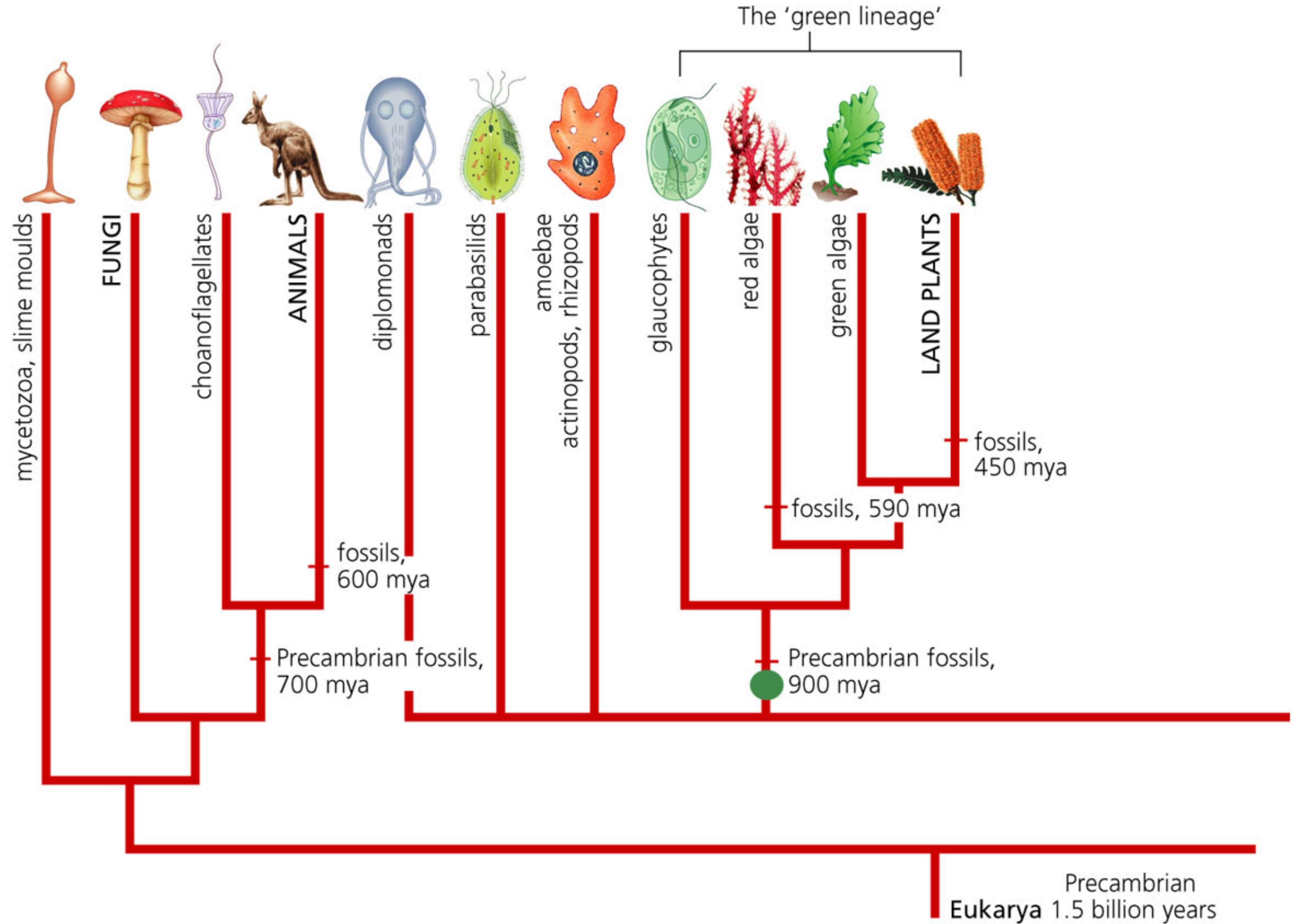
Levels of organisation and introduction to Eukarya



Dr Philip Button

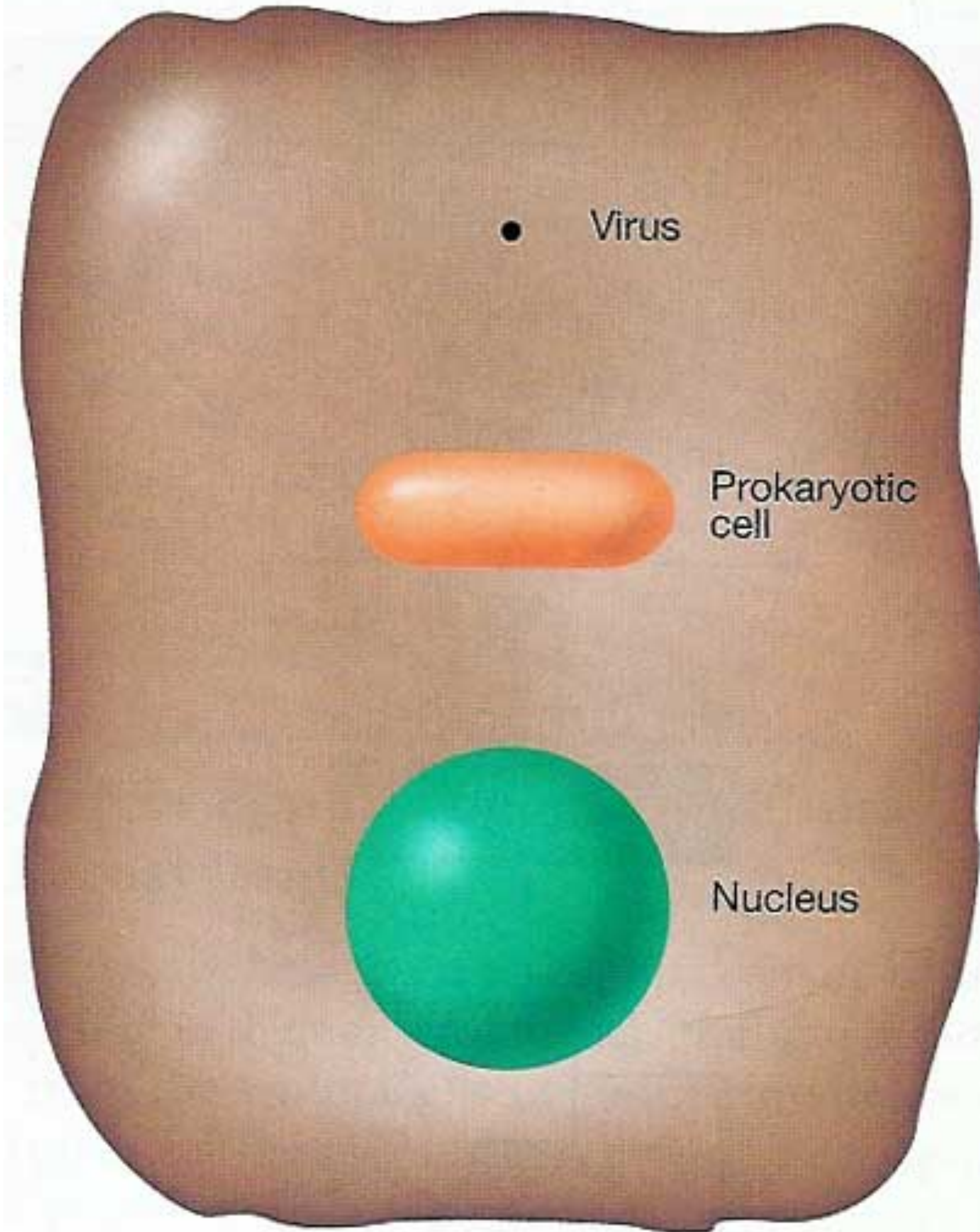






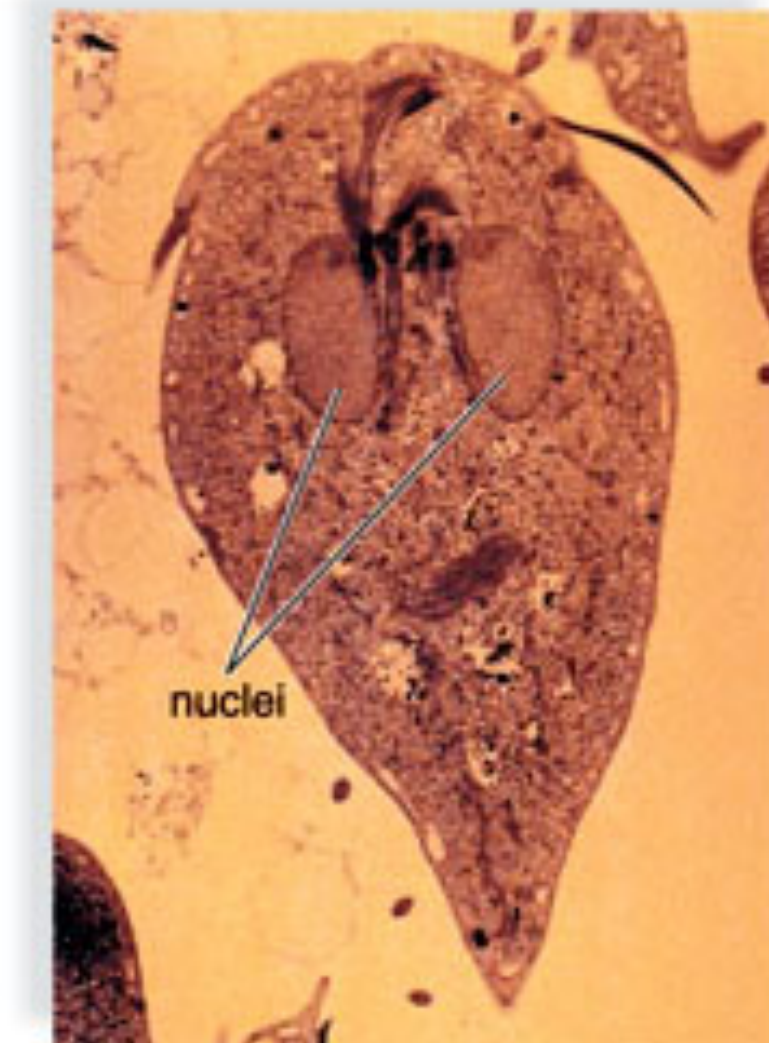
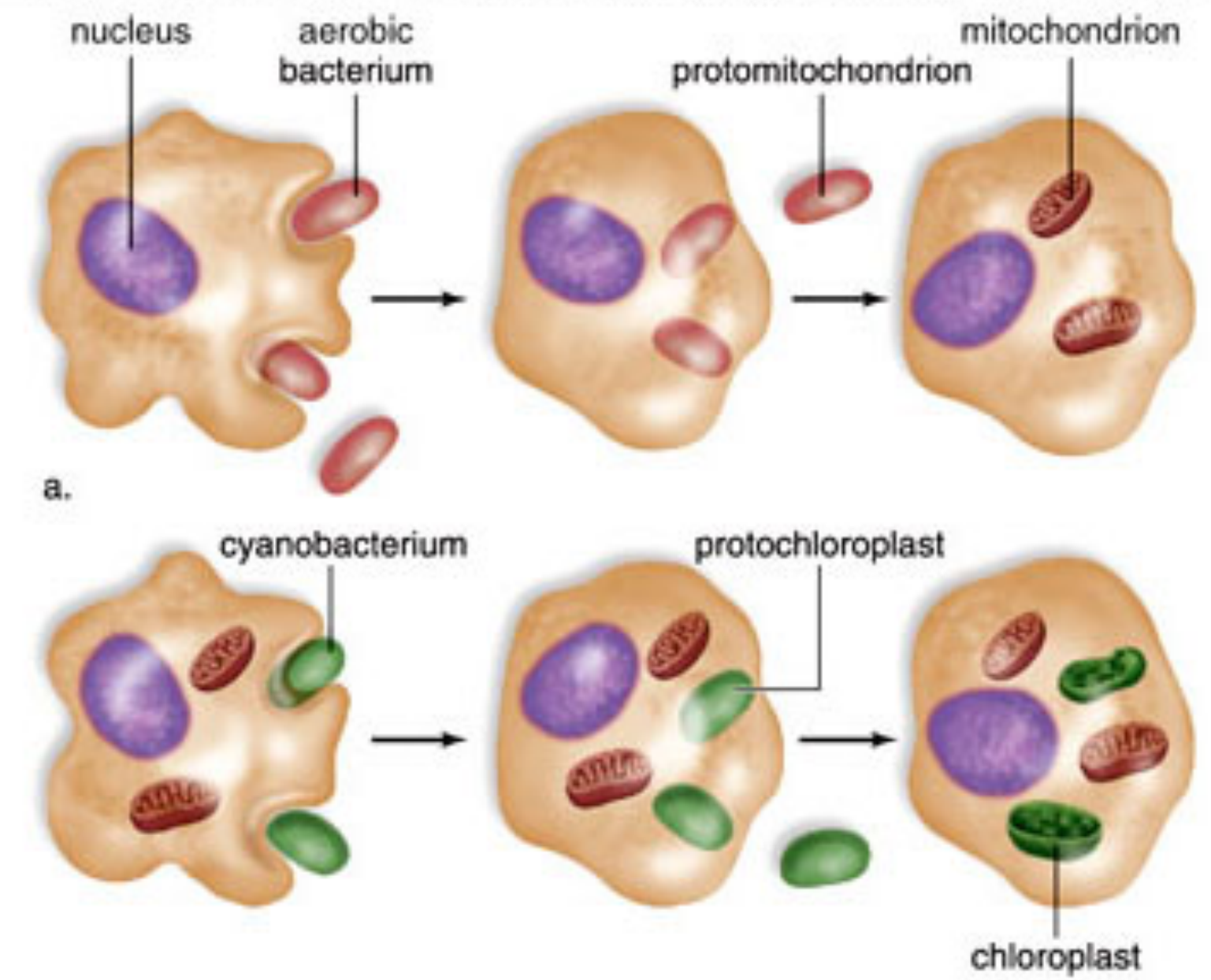


Typical eukaryotic cell



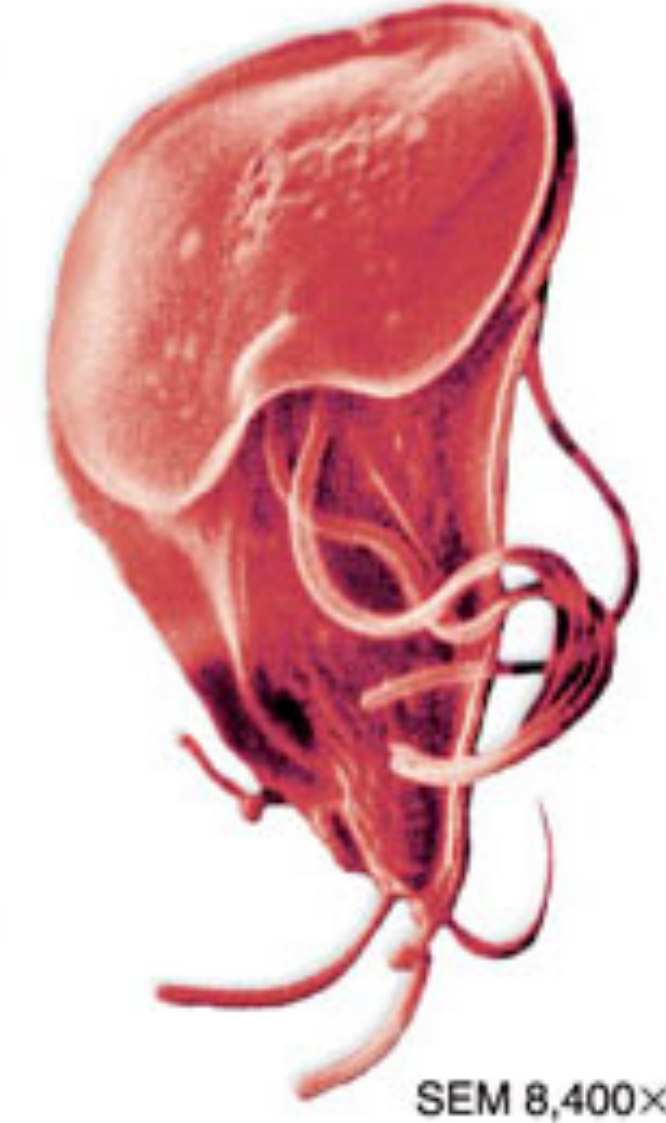
1000 nm (1 μ m)

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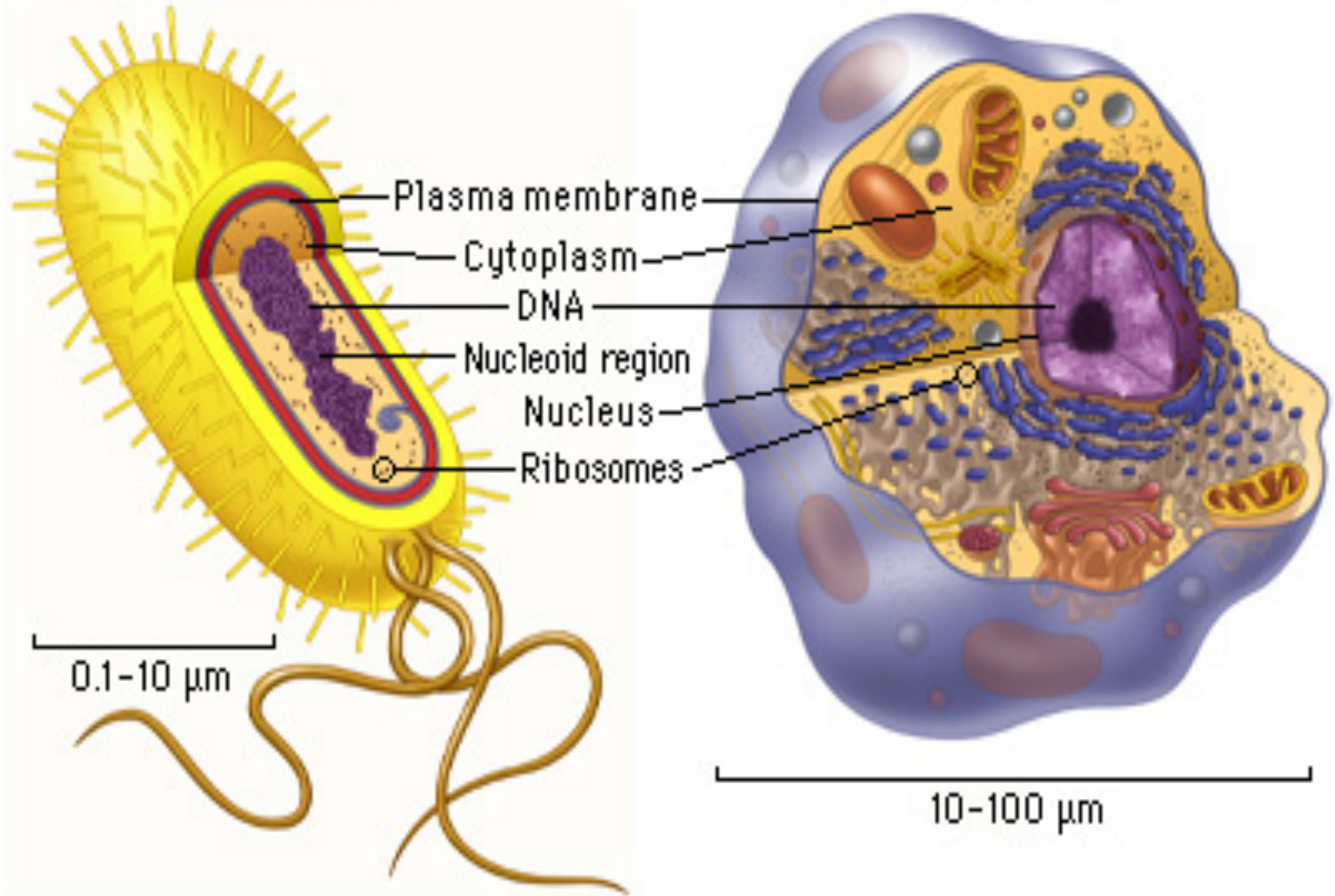
500 nm



SEM 8,400x

Prokaryotic cell

Eukaryotic cell



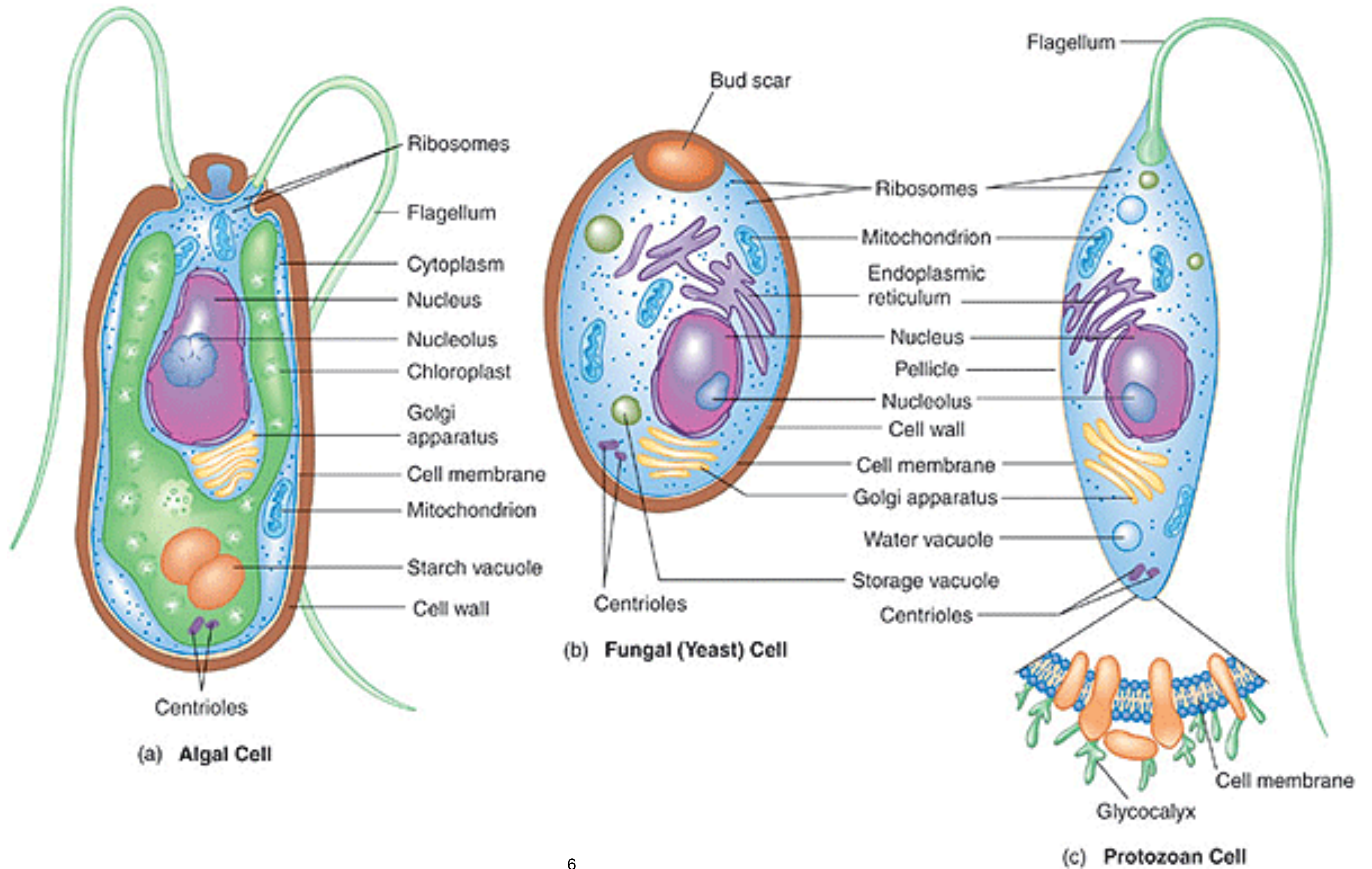
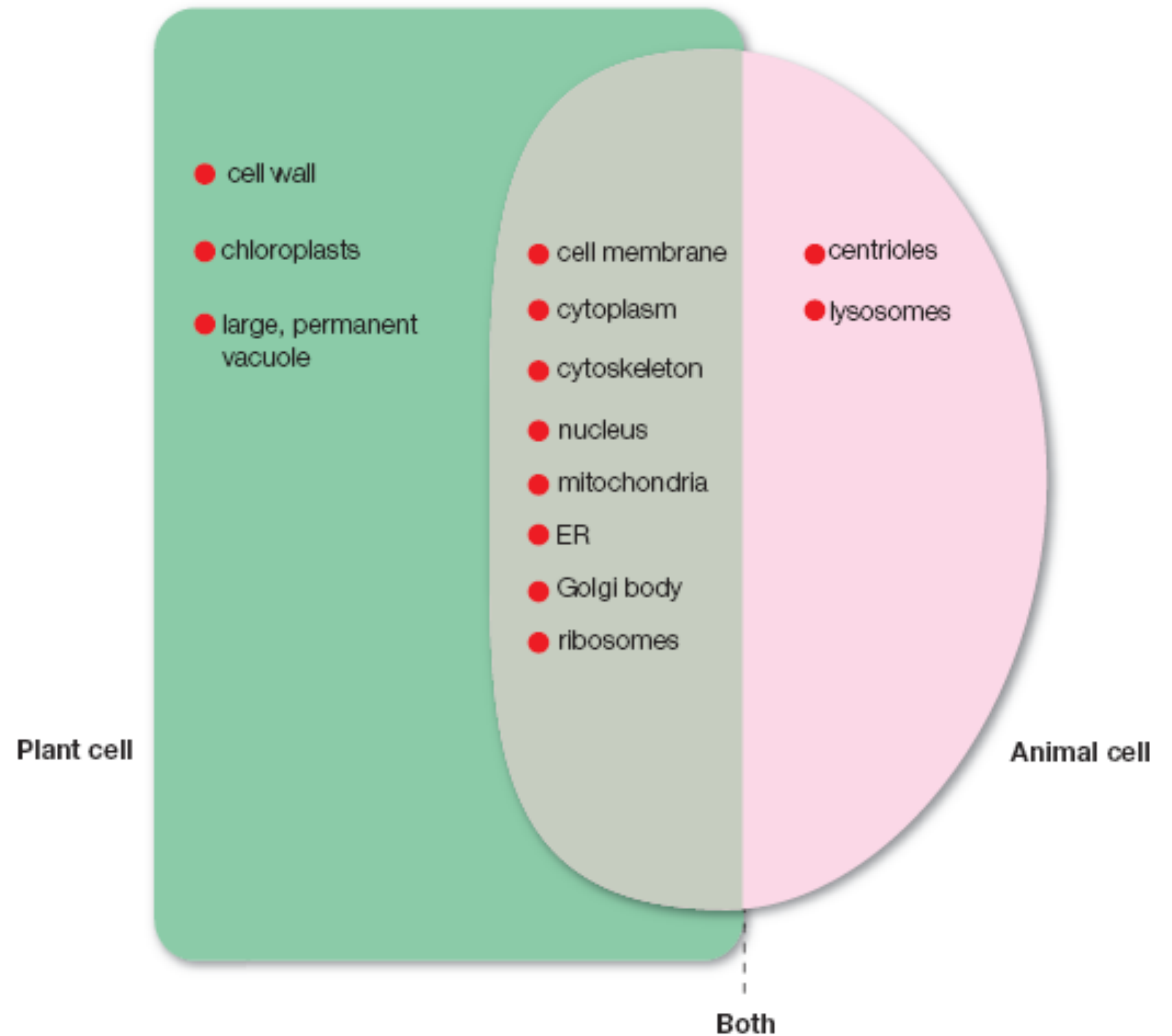


Figure 1.21

Summary of structures
found in plant and
animal cells



Level of organisation

CELL

Basic function unit of the body

TISSUE

Many cells with similar structure & function

ORGAN

Several tissues which perform a common function

SYSTEM

Several organs which perform the same general function

ORGANISM

Systems which work together to be a functioning animal





Level of organisation	Examples
* Multicellular organism	Plant, animal
Systems	Transport system, respiratory system
Organs	Heart, lungs, muscles, roots, stems, leaves
Tissues	Blood tissue, lung tissue, photosynthetic tissue
Cells	Nerve cell, blood cell, muscle cell, *unicellular organism
Organelles	Nucleus, chloroplast, mitochondria, ribosomes
Molecules	Water (H ₂ O), protein, sugar, lipid, chlorophyll
Atoms	Carbon (C), hydrogen (H), oxygen (O), nitrogen (N)

Diagrams:

